

DLP QUIZ-3

Week – 9

The Goal of computer vision is to extract meaning from the pixels. We see image through eyes. But computer sees only numbers.

Two types of information can be extracted from an image.

I. Semantic Information.

II. Metric Information.

Semantic Information such as :-

1. Object Recognition (Identify what is in image)
2. Object Deduction (Identify what is in image and locate it)
3. Image segmentation (Gives similar colors to similar kind of objects in the image)
4. Object tracking
5. Pose estimation
6. Activity recognition
7. Image captioning.

Metric Information includes :-

1. Depth estimation
2. Photo metric stereo
3. Structure from motion.

Challenges faced in these tasks :-

1. View point variation
2. Illumination
3. Scale
4. Deformation
5. Occlusion
6. Background Clutter
7. Motion and Intra class variation.

CNN vs MLP's

Naively using MLP's FC to classify causes parameter explosion. CNN overcomes it by,

1. Local connectivity
2. Parameter sharing
3. Pooling/sub sampling
4. Feature Abstraction
5. ReLU for non-linearity.

MLP have $m*n$ parameters to store and takes $O(m*n)$ runtime.

Whereas, CNN takes $k*n$ parameters, here k is less than m and takes $O(k*n)$ runtime.

With parameter sharing, CNN has only k parameters to store and takes $O(k*n)$ runtime.

CNN Convolution Layer work :-

It reduces the number of weight through local connectivity.

It provides spatial invariance through parameter sharing.

Hyper - parameters = Padding and Stride.

CNN Pooling Layer :-

It reduces the spatial size of representation, and to reduce the parameters and so the computation in the network.

Week – 10

Object Deduction :-

R-CNN aka Slow R-CNN

1. Input Image
2. Extract Region proposals (ROI from a proposal method)
3. Compute CNN features for each region.
4. Classify Images/Regions with SVM.
5. Apply bounding box regressor.

What's wrong with this? Log loss (CNN), hinge loss (SVM), Squared loss (Bounding box regressor)

Fast R-CNN :-

1. Input Image
2. CNN
3. ROI from a proposal method
4. ROI pooling
5. FC
6. Bounding box regressor + Softmax Classifier

So, Now we just need to calculate Log loss + smooth L1 loss and back propagate.

Still what's wrong? Out of box region proposal.

Faster R-CNN

1. Image
2. CNN
3. Feature Map
4. Region Proposal Net, its a fully convolutional network.
5. ROI pooling
6. From 4, Proposals, from 5 pooling, combining we get features.

Solely based on CNN. No external modules. Each step is end to end.

YOLO (You Only Look at Once)

Advantage of YOLO over R-CNN versions

Previous Approaches

1. A separate model for generating bounding boxes and for classification (more complicated model pipeline)
2. Need to run classification many times (expensive computation)
1. Looks at limited part of the image (lacks contextual information for detection)

YOLO

1. A single neural network for localization and for classification (less complicated pipeline)
2. Need to inference only once (efficient computation)
1. Looks at the entire image each time leading to less false positives (has contextual information for detection)

Overview :-

1. The image is split into $S * S$ cells.
2. For each cell, generate B bounding boxes.
3. For each bounding box, there are five predictions, x, y, w, h and confidence.
4. For each cell we also need to do object classification.

Output matrix size, $S * S (B*5 + C)$

Object deduction metric, mAP (Mean Average Precision),

How well the detector is able to localize and classify objects in the image. It's based on IoU, confusion matrix, Precision, Recall.

$$Precision = \frac{TP}{TP + FP}$$

$$Recall = \frac{TP}{TP + FN}$$

$$F_1 = \frac{2 \times precision \times recall}{precision + recall}$$

$$IoU = \frac{Area\ under\ Overlap}{Total\ Area}$$

Precision measures, How well you can find TP out of all predicted positives.

Recall measures, How well you can find TP out of all predictions.

Steps to calculate mAP score for Object Deduction :-

1. For each class, compute IoU, by using ground truth box.
2. Assign TP and FP. If $IoU > threshold$, TP else, FP, if there is a ground truth box with a prediction, then assign it as FP.
3. Compute Precision and Recall.
4. For each class create PR curve by varying threshold. Lower threshold leads to Higher FP.
5. Average Precision : The area under the curve is calculated by integrating the evenly separated area. This gives the average for each class.
6. mAP : Once AP for each class is computed, then compute average of all the class by adding them. It takes in to account that, not just objects are correctly deducted, but also the localization quality of bounding boxes.

YOLO makes far less background error, less likely to predict false positives, because IoU is very small with any ground truth label.

But it makes far more localization errors. Correct class, but IoU is small.

Week – 11

Monocular Depth Estimation methods :-

Its a challenging task. Depth cues like parallax is missing. Some cues that exist are vanishing points, object sizes, etc. These cues cannot be quantified through a rule based method.

Supervised method :-

Multi scale Deep Network :-

Depth map from single image using multi scale depth network, The entire network is divided into two parts.

1. Coarse Network
2. Fine Network

Coarse Network predicts the overall depth for the scene.

Fine Network refines the global prediction locally.

Since coarse network needs larger receptive field, it is more deep compared to Fine Network.

UNet:-

Depth map from single image using UNet architecture.

Entire network is divided into two parts.

1. Encoder Downsampling Network
2. Decoder Upsampling Network.

1. Uses Convolutional layer followed by max pooling to progressively reduce spatial dimension while increasing the number of channels. Captures high level features and context in the image.

2. Involves transposed convolutions (or upsampling layers) to increase the spatial dimensions of feature maps. Combines low-level features from the downsampling path with high-level features to produce detailed output depth maps.

Skip connections, directly connect downsampling path to the upsampling path. Preserve spatial information lost during downsampling, allowing more precise reconstructions in the output.

Loss function = MSE

Week – 12

Most common Image preprocessing tasks are,

1. Denoising
2. Super resolution
3. Deblurring

MPR Net (Multi stage Progressive Restoration Net)

Learns the contextualized features with encoder decoder architecture and then combines with high resolution branch that retains local information.

Multistage architecture progressively refines image quality across several stages.

Parallel branches extracts multi scale features for capturing both local and global information. Each stage outputs intermediate results passed to next for further enhancement.

The encoder decoder architecture is based on the standard UNet architecture. It incorporates CAB(Channel Attention Blocks) for enhanced feature extraction at multiple scales. Bilinear Upsampling followed by convolution is used to increase spatial recognition.

CAB :-

Channel Attention Map is essential for exploiting inter channel relationship of features. Channel Attention Block focuses on ‘What’ is meaningful given an input image. To compute the channel attention spatial dimension of input feature map is squeezed using average and max pooling, generating two different spatial context descriptors.

Both descriptors are then forwarded to shared network to produce the channel attention map.

Original Resolution Subnetwork:

Preserves fine details in image restoration.

Operates without downsampling, producing spatially enriched high resolution features.

Composed of multiple ORS blocks each containing CAB’s.

Cross Stage Feature Fusion :-

Its integrated between encoder-decoder and between the encoder-decoder and ORS net.

Features from one stage are refined using 1×1 convolution before passed to next stage

aggregation. Reduces information loss, enhances feature enrichment across stages and stabilizes network optimization process, facilitating addition of more stages.

SAM (Supervised Attention Module)

Introduced between stages to improve performance in multistage image restoration networks. It provides ground truth supervisory signals for progressive image restoration at each stage. It generates attention maps to suppress informative features, allowing only useful features to propagate next stage.

SRGAN (Photo Realistic Single Image Super Resolution Using a Generative Adversarial Network)

Its the first network in generating photo realistic images with $4\times$ upscaling factor.

Perpetual loss function. Combines adversarial loss and content loss.

Utilizes Generator G and Discriminator D to solve adversarial min-max problem where G aims to fool D into classifying generated image as real. G generates realistic image while D tires to distinguish between super resolved and real images, encouraging perpetually superior solutions over pixel wise error methods like MSE.